

Demonstrating TapID for Rapid Touch Interaction on Surfaces in Virtual Reality for Productivity Scenarios

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Figure 1: We demonstrate multiple applications that bring rapid touch interaction to Virtual Reality, which is enabled by our integration of VR headset-based hand pose tracking and touch detection on a table. To detect touches, we demonstrate our prototype TapID, which is a wrist-worn sensor device that registers surface taps and identifies the tapping finger. Using this recognition, TapID triggers the corresponding UI elements in VR. (A) A user is wearing a VR headset and two TapID bands to (B) interact with surface widgets through touch, e.g., for text input (left), web browsing (right), or (C) document authoring in a spatial setting.

ABSTRACT

We demonstrate a novel approach to bring quick touch interaction on surfaces to Virtual Reality, which is a challenge for current camera-based VR headsets that support free-hand mid-air interaction or physical hand-held controllers for input. In our approach, we use our wrist-worn prototype TapID to complement the optical hand tracking from VR headsets with inertial sensing to detect touch events on surfaces, which establishes the same interaction modality that is present on today's phones and tablets. Each TapID band integrates a pair of inertial sensors in a flexible strap, from whose signals TapID reliably detects surface touch events and identifies the finger used for touch. This event detection is then fused with the optically tracked hand poses to trigger input in VR. Our demonstration comprises a series of VR applications, including UI control in word processors, web browsers, and document editors.

These applications showcase the beneficial use of rapid tapping, typing, and surface gestures in Virtual Reality.

CCS CONCEPTS

• **Human-centered computing** → **Human computer Interaction (HCI)**; **Human Computer Interaction (HCI)**.

KEYWORDS

touch input, VR, productivity, inertial sensing, neural network

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1 INTRODUCTION

Recent Mixed Reality systems, Augmented Reality and Virtual Reality systems alike, move towards detecting input from recognized hand poses (e.g., Microsoft HoloLens 2, Oculus Quest 2), which supersedes the previously established interaction model using hand-held controllers for input. In AR systems, hand tracking is suitable to

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support the work on physical objects, which allows the user to hold onto something and prevents fatigue during use. In VR systems in contrast, all content is intangible and gesture-based interaction can become burdensome beyond quick interactions. A large set of research efforts have therefore focused on introducing haptic-feedback mechanisms for general-purpose interaction in VR (e.g., active devices [2] or passive haptics [1]).

We demonstrate a novel approach to interaction in VR and move all *direct* input in VR to passive surfaces [3]. Compared to mid-air interaction, touch interaction on surfaces provides users with an opportunity to rest their arms on between interactions while simultaneously offering physical support during prolonged interactions. At the same time, the surface provides haptic feedback alongside a frame of reference for proprioception, affording quick interaction.

The challenge of using input on surfaces in VR systems is the precise detection of touch events; because hands are tracked from the head-mounted systems, the single vantage point can lead to inaccuracies of depth sensing and thus touch/no-touch discrimination and contact positioning [5].

In our demonstration, we introduce reliable and quick touch detection with our prototype bands TapID, which users wear around both wrists. Through built-in inertial sensors, TapID does not just detect touch events on passive surfaces but additionally identifies the finger used for touch. TapID thus complements the headset’s optical tracking of hands and fingers, with which we fuse touch events and thus enable the seamless and reliable transfer of touch applications and interaction modalities known from tablets and phones.

Our demonstration is shown in Figure 1. The user is interacting through freehand motions in VR, while two TapID bands detect finger touches on the table in front. The inertial events registered by TapID are streamed to a PC for tap classification and finger identification. Our VR system then fuses hand pose estimates from the headset with TapID’s events. Figure 1b shows an application that benefits from our approach: a keyboard that affords rapid and reliable text entry using all ten fingers. Upon typing on the keyboard, our system forwards all key input to the operating system for text processing, so users can type similar to how they would interact on a regular touch tablet. The text editor also supports functionality users would expect from tablets, such as tapping to place the cursor, double-tapping to select a word, tool buttons, etc.

2 IMPLEMENTATION

TapID’s hardware implementation is centered around a DA14695 System-on-a-Chip (Dialog Semiconductor) as shown in Figure 2, a 32-bit microcontroller (ARM Cortex M33) that is connected to two low-power accelerometers (LIS2DH, STMicroelectronics), which are mounted on a flexible PCB. All electronics are embedded in an elastic silicone-based strap. We designed the form factor of the board to be comparable to that of a smartwatch. Each TapID band is tethered to a host PC for power and serial communication purposes.

The accelerometers on the flexible PCB mechanically couple to the user’s skin and bones through the strap and pick up body-coupled vibration signals, sampling accelerations at 1.3 kHz. During each tap event, the inertial sensors register slightly different

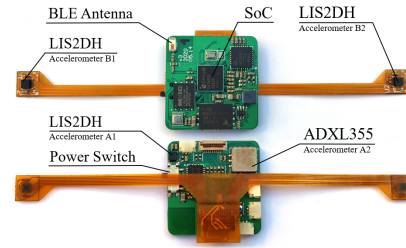


Figure 2: The main PCB features a microcontroller that reads out the two low-power IMUs that are mounted on the ends of a flexible PCB. The flex PCB connects to the main PCB through an extension connector. For comparison in our evaluation, two additional IMUs are mounted on the main PCB.

vibration signals due to the propagation path through the user’s hand depending on the finger they originate from. First, tap events are recognized from the sharp spikes in the raw signals of the accelerometers. Second, once a tap has been detected, we analyze the differences between the inertial signals in a machine-learning pipeline to identify the tapping finger. We feed the acceleration data from both sensors as input into a multi-layer feed forward 1D-convolutional neural network classifier following a VGG-style architecture. The network is trained using cross-entropy loss on a dataset we recorded from 18 participants who delivered taps on a table surface with different fingers. In an evaluation on those records, we found that our approach can detect taps with an accuracy of $F_1 = 0.997$ and identifies the correct finger with an accuracy of $F_1 = 0.93$ in a within-user evaluation setting. When trained on cross-user data, the accuracy of finger detection is $F_1 = 0.87$ and increases to $F_1 = 0.91$ after fine-tuning with 10 user-specific samples.

3 DEMONSTRATION AND APPLICATIONS

We demonstrate several applications that showcase our unique combination of rapid tap interaction in VR through TapID. In all demonstrations, the user is situated in VR and surrounded by virtual content reminiscent of desktop user-interface affordances. These include photos, document editors, web browsers, and word processors as shown in Figure 1B. We implemented several *tapping-based virtual input devices* that the user can open and position by tapping at the desired location on the surface with specific fingers. For instance, the left pinky finger summons arrow keys, the right pinky finger a numeric keypad and the thumb summons a simple keyboard between both hands. In the following, we describe the specific prototypes that we built based on this principle. Screenshots of the four prototypes can be seen in Figure 3 (please also see the supplementary video for a complete demonstration).

Photo sorting: Common on touch devices, our *Photo sorting* app (Figure 3.A) uses TapID’s reliable tap detection to trigger drag events for moving and resizing pictures. A command input menu enables step-wise resizing and rotating images. With our approach, we can easily assign these commands to each finger.

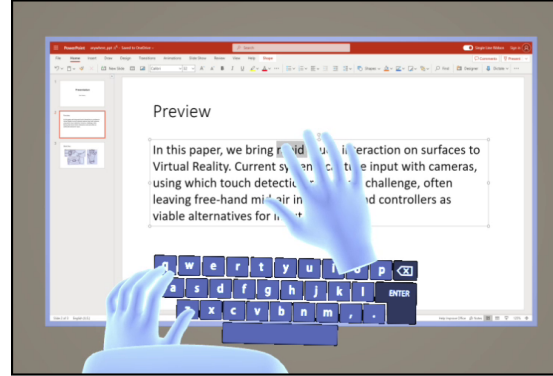
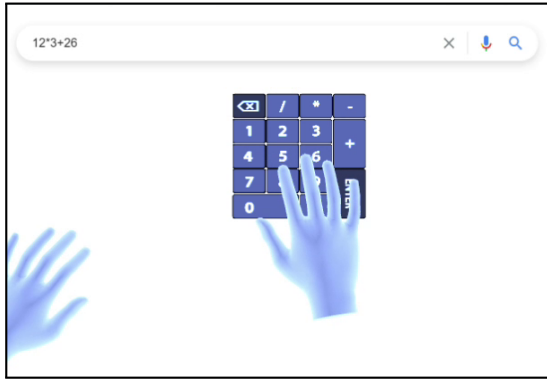
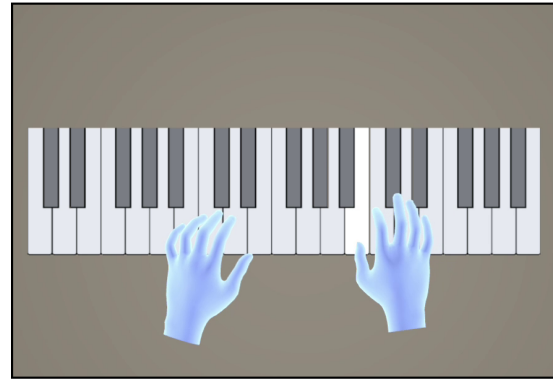
A: Photo sorting**B: Document editing****C: Number input****D: Piano**

Figure 3: Our demo applications showcase TapID’s integration with VR scenes and hand-based interaction using different surface widgets. (A) dragging and resizing photos including a per-finger command input widget below the left hand, (B) editing rich-text documents with keyboard input including text cursor navigation, (C) a numeric keypad for inputting numbers and operations, and (D) virtual piano keys for precise input.

Document editing: We designed a simple keyboard input widget to add text to documents (Figure 3.B), complementing the use of existing document authoring tools (e.g., Microsoft Word, PowerPoint). Words can be selected through double-tap, e.g., to change their format. Additionally, our arrow key virtual input (summoned with left pinky finger) aids in quickly adjusting the text cursor’s desired location. This demonstrates our support of conventional touch capabilities, seamlessly integrating our finger-specific tapping-based virtual input devices.

Number input: Users can open a virtual numeric keypad widget to type in numbers and operations (Figure 3.C). This widget conveniently fits below the hand, so that users can quickly and reliably input numbers using all five fingers of one hand, e.g., to use it as calculator input in a user interface.

Piano: Our final demonstration is a virtual piano app (Figure 3.D). This app optimally leverages TapID for temporally precise input, enabling users to hit piano keys using any finger. The combination of visual feedback in VR, the passive haptic feedback of the surface, and the low-enough latency of our device uniquely enables playing this virtual instrument while preserving the exact beat of the user’s tapping input.

4 CONCLUSIONS

We presented a demonstration of a novel approach of input in Virtual Reality that fuses rapid tap input on passive surfaces with hand pose input tracked by VR headsets. Touch-input detection is provided by our prototype TapID, a wearable sensing device that reliably detects tap events and identifies which finger has been used for the tap. This allows VR systems that track the user’s hands to add an interaction modality that is currently absent, which is touch input in Virtual Reality. In our demonstration, we use this input for rapid surface interaction through disambiguating which finger causes an action and fusing this information with hand poses to trigger input events in the VR application. Taken together, our approach is a viable complement for existing VR systems that allows the familiar input modality that is touch input to transfer to spatial VR systems, potentially even during everyday scenarios. Our approach can also serve as the basis for prolonged interaction, such as in interactivity scenarios that involve touch and pen interaction [4]. For full details of our prototype TapID and its evaluation, please see the paper [3].

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